

Kinematic Predictor for Agility Run and Short Sprint Acceleration Run in Field Hockey Trainees

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ABSTRACT

The purpose of the present study was to determine the relationship between Kinematic parameter (Peak Velocity during Countermovement Jump) with Agility Run and Short Sprint Acceleration Run. A total of 124 (66 males and 58 females) hockey trainees, participated in this study. After thorough warm-up, participants were first made to perform CMJ on Force Platform and then were asked to run for 5-meter Short Sprint Acceleration Run as well as 6x10m Shuttle Run. Correlation of Peak Velocity with Short Sprint Acceleration Run and Agility Run was found as $r = -0.542$ and $r = -0.457$ respectively for all subjects, while male hockey trainees demonstrated the values of $r = -0.794$ and -0.699 , whereas the female hockey trainees indicated the values of $r = -0.632$ and $r = -0.575$ with Short Sprint Acceleration Run and Agility Run respectively. The present study concluded that Peak Velocity had linear correlation with Short Sprint Acceleration Run as well as Agility Run. This could be a fruitful result for coaches as well as players to inculcate Vertical Jump regime in their training program so as to improve the performance of the players.

Keywords: Field Hockey, Kinematics, Peak Velocity, Counter movement Jump, Agility, Acceleration Run

INTRODUCTION

Most of the field games demanded movements, like sprinting, jumping, throwing, and change of direction and other related activities. These performances were common in, for example, track and field, rugby, football, basketball, field hockey, etc. and shared

the involvement of large numbers of muscle tissues over more than one joint and those must be recruited at appropriate times and intensities to maximize speed (Markström JL & Olsson CJ. 2013). In fact, it had been proposed (Cronin J & Sleivert G. 2005) that multi-joint

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exercises were preferred as training and testing modalities considering the nature of athletic performance. However, it had also been shown that players from different sports elicited different characteristics (Laffaye et al., 2007). More specifically, when elite athletes were evaluated, they were characterized with an ability to develop high amount of force quicker than other groups during drop jumps (DJs). Further, elite sprinters had proven to generate better peak velocities and jump heights when performing loaded counter movement jumps (CMJs) (Comfort et al., 2012).

Sprint performance was arguably the most needed fitness feature for success in team sports, wherein the capability to accelerate more quickly and outrun an opponent was vital (Bangsbo et al., 2006). A faster athlete had an evident advantage during decisive part of a match because he or she had a higher chance of getting to a ball. During team sports, the most critical part in a game were often determined by intermittent high-intensity sprints those rarely lasted 01–22 meters (Wither, 1982) but when combined might account for a significant portion of the match, making up approximately 11% of game time in soccer—or a 10- to 15-meters sprint every 90seconds (Meir et al., 2001), with similar findings also reported in rugby league (Cunniffe et al., 2009), and rugby union (Spiteri et al., 2014). The basic criterion of effective

sprinting velocity was to develop the highest possible ground reaction force in the shortest time possible during the contact phase of single running stride development of force as a result of relationship among eccentric and concentric muscular contractions.

Agility and change of direction (COD) were common athletic maneuvers requiring athletes to own an aggregate of physical, technical, and tactical attributes to steer clear off or pursue opponents. Agility including COD involved a combination of multiple strength components (Peterson et al., 2006). The relationship between estimates of strength, COD, and agility performance was not entirely linear (Nimphius et al., 2010), still, coaching techniques which aimed on the development of leg strength and power (Keiner et al., 2014) appeared to produce advancement in COD performance. Likewise, exploration studies examining beyond across-sectional relationship, had delved into the relationship between changes in strength (1 RM squat), and changes in COD performance over time and had indicated that moderate to strong correlations existed (Hori et al., 2009). The inconsistent relationship determined among strength and COD functionality might be attributed to elements which included variations in the level of athlete, gender, training, age, and more importantly the strength assessment used

(Nicholson et al., 2021).

The aim of this research was to determine the connection between Kinematic parameter (Peak Velocity during Counter movement Vertical Jump (CMJ) with Short Sprint Acceleration Run and Agility Run in field hockey trainees. It was envisaged that the findings would determine the contribution of each strength component to functional Short Sprint and Agility performances in field sport players. The purpose was specifically aimed at our unflawed understanding of the specific aspects of “velocity” that might be associated with sudden short sprinting and agility performances and also to improve the underpinning physical qualities looked-for to muster the sports along with other skills.

MATERIAL AND METHODS

A total of 124 field hockey trainees including 66 males ($n_m = 66$) and 58 females ($n_f = 58$) were taken. The mean age and mean weight of the participant's were 15.58 ± 1.23 years and 60.34 ± 5.80 Kg respectively. Among the males, 21 were stickers, 28 were mid fielder and 17 were defender while females were 17, 21 and 20 respectively. All the players had at least 1 year of training age at the time of study. It was designed to find out the effect of Kinematic parameter (Peak Velocity during CMJ) on Short Sprint Acceleration Run and Agility Run in

selected field hockey trainees.

The Force Platform (KISTLER 9281EA, Switzerland) was used to measure the Peak Velocity during Counter movement Vertical Jump. Force platform is a rectangular, 0.4 meter x 0.6 meter size with piezoelectric sensor attached at each corner to give an electrical output that is proportional to the force on the plate. Electronic Running Timer (Swiftspeed light SL OPT-0101, Australia) was used to record the time at various distances during Shuttle Run and 05 meter distance for Short Sprint Acceleration Run. Electronic running timer (popularly known as electronic timing gates) is a device which makes an invisible beam of light between photocell and reflector, which when crossed by athlete, the time got recorded in recording unit.

Procedure

The participants, in the presences of their coaches, were instructed thoroughly regarding the procedure prior to the chosen tests. Field hockey trainees were required to wear proper sports clothing and shoes. Five minutes warm-up session was given to them before the commencement of the tests. They were further instructed to perform 6x10 meter Shuttle Run and Counter movement Vertical Jump (CMJ). For 6x10 meter Shuttle Run Test, ten-meter distance was marked by two parallel line of 1.5-meter

width. The participant stood behind the starting line at one end and on the signal, he had to run as fast as possible to the other end of the marked track and touch the line with foot, turn and back to the starting line, touch it with foot again. They were made to understand to turn and repeat it for a total of six times. Gates of Speed light running timer were placed at a distance of 5 meter from the starting line, so that time taken to first 5 meter (Short Sprint Acceleration Run) was recorded. Shuttle Run was followed by CMJ on

Force Platform (figure 1) which was set at the frequency of 200Hz. While performing Vertical Jump, selected field hockey trainees were instructed to keep hands on their respective waists so as to minimize the use of arms swing for jump and also to maintain the uniformity of the jump. The data from running timer was recorded in the recording unit connected to Gates via cable and RJ connector. Times taken for first 5 meter by the participants were recorded directly from the recording unit.

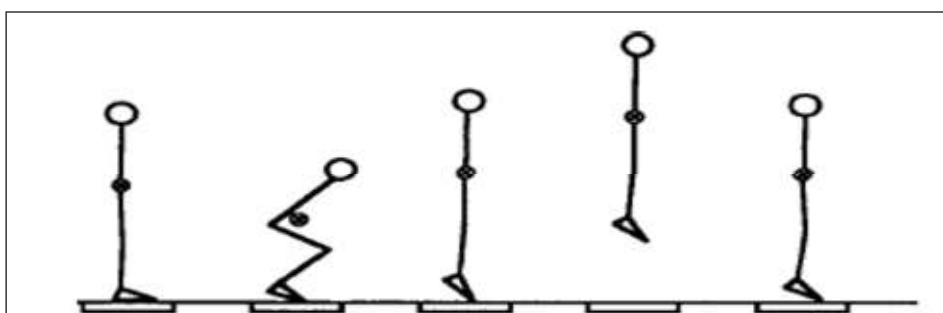


Figure 1: Sequence of actions in a countermovement vertical jump. ⊗ - indicate Jumper's center of mass (Linthorne, N. P., 2001)

Statistical analysis was performed using SPSS software (version 22.0. SPSS. Inc., II.). Correlation between dependent variables (Short Sprint Acceleration Run and Agility Run) and independent variable (Peak Velocity during CMJ) was detected using Pearson's Correlation. The correlation amongst field hockey trainees (N=124) as well as between male and female participants were calculated separately. Level of significance was kept at $p \leq 0.05$, 'r-values' were interpreted as

weak (≤ 0.39), moderate ($0.40 \leq r \leq 0.69$) or strong (≥ 0.70).

RESULTS

Table 1 showed the mean and standard deviation of the selected variables for total subjects (N=124), and male (nm=66) and female (nf = 58) field hockey trainees. It was found that female trainees generated higher Peak Velocity while males had quicker time for Short Sprint Acceleration Run as well as Agility Run.

Table-1 Descriptive Statistics of the Selected Parameters			
	Variables	N	Mean $\pm$ SD
Total	Peak Velocity (m/s)	124	02.18 $\pm$ 0.65
	Short sprint – 5meter (s)		01.90 $\pm$ 0.11
	Agility – 6x10m (s)		17.67 $\pm$ 1.11
Male	Peak Velocity (m/s)	66	02.16 $\pm$ 0.15
	Short sprint – 5meter (s)		01.88 $\pm$ 0.09
	Agility – 6x10m (s)		17.12 $\pm$ 0.52
Female	Peak Velocity (m/s)	58	02.21 $\pm$ 0.95
	Short sprint – 5meter (s)		01.93 $\pm$ 0.12
	Agility – 6x10m (s)		18.30 $\pm$ 1.28
Where, m/s = meter per second, s = second			

Table 2 clearly indicated that there was significant relationship of peak velocity with Short Sprint Acceleration Run and Agility Run at 0.01 level of significance

for overall subjects and also among male and female trainees separately as well. The same has been shown separately in figure 2.

Table-2 Correlation between Dependent Variables (Short Sprint and Agility Run) and Independent variable (Peak Velocity during CMJ)

		Short Sprint	Agility
Peak Velocity	Total	-0.542**	-0.457**
	Male	-0.734**	-0.699**
	Female	-0.632**	-0.575**

**significant at the 0.01 level (2-tailed)

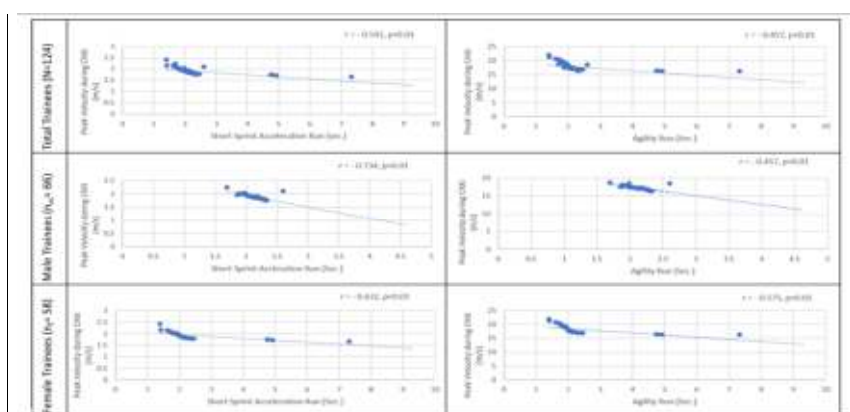


Figure 2: Relationship of Peak velocity with Short Sprint Acceleration Run and Agility Run in field hockey trainees

DISCUSSION

The aim of the present study was to determine whether Peak Velocity during Counter movement Jump governed the performances of selected field hockey trainees on Agility Run and Short Sprint Acceleration Run. Such information would give clear directions for trainers and athletes in the development of training programs and also in making the provisions of applicable feedback when coaching and guiding players for better acceleration. The results of the current research clearly showed that a peak velocity of CMJ differentiated individuals/field hockey trainees who were different in terms of their early acceleration skills as well as proficiencies.

In field sports and track events, the player's acceleration had been recommended to be an essential determinant of performance (Haugen et al., 2019), and acceleration was defined as the rate of change of velocity, which was the independent variable in the current study. The findings of this investigation depicted the significant correlation of Peak Velocity during CMJ with Short Sprint Acceleration Run as well as with Agility Run. Peak Velocity was found to have moderate correlation with Short Sprint Acceleration Run ($r = -0.542$) and Agility Run ($r = -0.457$) in case of field hockey trainees ($N = 124$) taken together as one group.

In particular, males ($nm = 66$) had strong correlation value of Peak Velocity with Short Sprint Acceleration Run as $r = -0.734$ and Agility Run as $r = -0.699$. In case of female hockey trainees ($nf = 58$), correlation among Peak Velocity and Short Sprint Acceleration Run was found to be strong ($r = -0.632$) while the correlation of Peak Velocity with Agility Run was moderate ($r = -0.575$). The negative values of the correlation indicated improvement in the performance, if peak velocity during CMJ increased, the time taken to complete Short Sprint and Agility Run decreased. Males showed stronger correlation than female trainees among all the selected parameters. The difference in the association amongst gender might be due of the technique used or the exposure of such movements to different athletes during their practice session. This gender difference might also be related to the ability of applying large force quickly in male athletes as compared to females. This might be the scope of the future research.

There are number of scientific studies which supported the findings of our current investigation. Ates & Cetin 2017 reported in a meta-analysis that short-sprint performance could be improved by combining secondary and tertiary training methods. Combined specific training methods which includes CMJ training as well along with other

methods also showed modest significant enhancement in short-sprint performance at 0–5 m and 0–10 m. Similar noteworthy findings also were supported by Haugen et al (2019) in an integration of scientific and best applied literature. A significant negative relationship was found among CMJ and 10-m and 30-m sprint values ($r = -0.59$) and ($r = -0.63$) respectively (Vescovi & McGuigan, 2008). Statistically significant results were also found between reactive force, regarding jumping, flat sprint and sprint in different directions in a study related to soccer, basketball, Australian soccer and tennis athletes, studying the relationship between falling and leaping and 8 m sprint performances in straight and different directions (Braz et al., 2017). Researcher also established a significant relationship between the two vertical jumps (CMJ and SJ) and the relationship between sprint velocities and variables (Priya et al., 2021). At the end of the study, a significant negative correlation ($r = -0.69$, $p < 0.05$) was noted between CMJ and 10 m sprint. A stronger correlation was found in college and high school women, and it was determined that in longer distances (27.4 and 36.6 m) with CMJ was observed than short distances (9.1 and 18.3 m). (McFarland et al., 2016).

Agility in the current study had indicated that moderate correlations were established for trainees of both sexes taken together as a group ($r = -0.457$) and

female trainees ($r = -0.575$), and strong correlation was also discerned among male trainees ($r = -0.699$) with Peak Velocity during CMJ. Priya et al (2021) undertook study to find out the relationship between explosive strength (including velocity as a component) and agility among college level footballers, and it was found that significant moderate negative correlation ($r = -0.536$) had been perceived between chosen variables. Similar kind of strong correlation between CMJ and T-test but weak correlation in case with the other agility test (for e.g., Pro agility test) was documented (Vescovi & McGuigan, 2008). Corresponding result was depicted on women and men playing in the 2nd league university soccer (McBride et al., 2009). In a study of the association among two different jumping analysis (active and squat jumping) and sprint and directional change velocity abilities, a strong negative relationship ($r = -0.76$) was identified among CMJ and T-Test only in female soccer players, whereas there was no significant relation among CMJ and agility tests in male players. In another research work (Cronin & Hansen, 2005), it was proposed that there has been a strong correlation ($r = -0.53$) among the CMJ and the Zig Zag Test in the study of Brazilian elite soccer players. The findings are in parallel with the literature. Coaches can use the CMJ test, particularly once examining the

maximal in field game players. Some studies had emphasized that high stride frequency was important for fast acceleration (Vescovi & McGuigan, 2008). Our data gave empirical support to this finding. In particular, the present data showed that those athletes who were able to generate higher Peak Velocity during CMJ had depicted quicker time for Short Sprint Acceleration Run. Similarly, we advocated that a focus on reduced ground contact time need to be an important consideration of any sprint acceleration training program. Evidence indicated that a rise in lower limb musculotendinous stiffness may allow athletes to attain quicker ground contact times during running and sprinting (Vescovi & McGuigan, 2008).

The ability to accelerate effectively, and thereby out-accelerate an opponent, over the first few meters might be more crucial to attain success in those sports activities than top running velocity. Athletes, therefore, dedicated their massive quantities of training time and efforts toward sharpening their overall performances inside the pivotal range. An assortment of research had investigated the connection between strength and sprint performance and determined that, in general, stronger athletes performed better during sprint performances.

CONCLUSION

It was concluded that Peak Velocity

during CMJ had linear correlations with Short Sprint Acceleration Run as well as Agility Run in field hockey trainees. The results of this study illustrated the importance of developing high levels of lower-body strength and quickness (velocity) to enhance acceleration run and agility which are prerequisites for field hockey game, where quicker athletes had demonstrated superior sprint and agility performances. As with other researches, this study had its limitations. Since only young hockey trainees were included in the study, the results of this research do not apply to the entire hockey playing population. Future studies should also consider different levels of competition, age groups, gender and game positions when examining the relationship.

Applicable Remarks

Coaches as well as strength and conditioning experts should ensure that youth players develop counter movement jump velocity, once athletes are technically proficient, which should result in improvements in both Short Sprint Acceleration Run and Agility Run. Counter movement Jump should be improved as part of a periodized training program, ensuring that technical proficiency in all performances was enhanced.

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